GOVERNMENT OF WESTERN AUSTRALIA

GSWA Geochronology Record

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Department of Mines, Industry Regulation and Safety

242442.1.pdf *Geochronology Record 1931*

242442.1: feldspathic gneiss, Passchendeele farm

(Thundelarra Supersuite; Youanmi Terrane, Yilgarn Craton)

SS Romano, MTD Wingate, Y LU, IOH Fielding and TJ Ivanic

Location and sampling

PERENJORI SH50-06, CARNAMAH 2038 Zone 50, MGA 398957E 6728666N
Warox Site TJIROT190061 Sampled on 15 May 2019

This sample was collected from outcrop along the Carnamah – Perenjori Road, about 11.4 km southeast of Three Springs talc mine, 5.1 km east-northeast of Inering Hill, and 3.6 km west of Passchendeele farm.

Geological context

The unit sampled is a feldspathic gneiss (Fig. 1) of the Thundelarra Supersuite (Ivanic et al., 2022). The 3010–2930 Ma Thundelarra Supersuite consists mainly of migmatitic tonalite to granodiorite gneiss, and is the oldest basement unit in the Youanmi Terrane, coeval with the oldest recognized felsic metavolcanic rocks of the Southern Cross Supergroup, including the Wongan, Madoonga, Gossan Hill, Mount Gibson, Honman, and Annabelle Volcanics Formations (Van Kranendonk et al., 2013; Ivanic et al., 2022). Felsic metavolcanic rocks 44–49 km to the northeast, in the Koolanooka greenstone belt, yielded igneous crystallization ages of 3002 ± 4 Ma (GSWA 242672, Wingate et al., 2021b) and 2961 ± 4 Ma (GSWA 242652, Wingate et al., 2021a). A metamonzogranite sampled about 69 km to the northeast yielded an igneous crystallization age of 2946 ± 4 Ma (GSWA 219311, Wingate et al., 2020).

Figure 1. Hand specimen image of sample 242442: feldspathic gneiss, Passchendeele farm

Petrographic description

The sample is a foliated, fine to medium-grained feldspathic gneiss (Fig. 1), consisting of about 43% plagioclase, 32% quartz, 17% K-feldspar, 6–7% biotite, 2% myrmekite, <1% magnetite and accessory allanite, apatite and zircon. Plagioclase is subhedral to anhedral, equant, up to 3 mm in size, and variably sericitized. Quartz is anhedral, irregular, up to 3 mm in size, and exhibits undulose extinction and minor subgraining. The larger quartz grains have coarsely intergrown margins with adjacent feldspar. K-feldspar is microperthitic and forms simply twinned phenocrysts up to at least 9 mm long, as well as anhedral, interstitial grains up to 1 mm in size, some of which exhibit microcline twinning. Curved boundaries between many quartz and feldspar grains indicate grain boundary adjustment during metamorphism and deformation. Lobes of myrmekite up to 1 mm long have partially replaced some K-feldspar grains. Biotite forms slender, subhedral laths up to

1.5 mm long and small, aligned decussate aggregates that define the foliation. Equant subhedral magnetite is up to 0.7 mm across, and metamict allanite is subhedral to euhedral and up to 0.8 mm in diameter. Apatite is subhedral, tabular and up to 0.4 mm. Deformation produced the biotite foliation and strained quartz and feldspar and was followed by recovery at moderate metamorphic grade.

Zircon morphology

Zircons isolated from this sample are pale brown to dark brown or opaque, subhedral to euhedral, and variably rounded. The crystals are up to 400 µm long, and equant to elongate, with aspect ratios up to 6:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and many crystals contain high-U, metamict zones and inclusions. A CL image of representative zircons is shown in Figure 2.

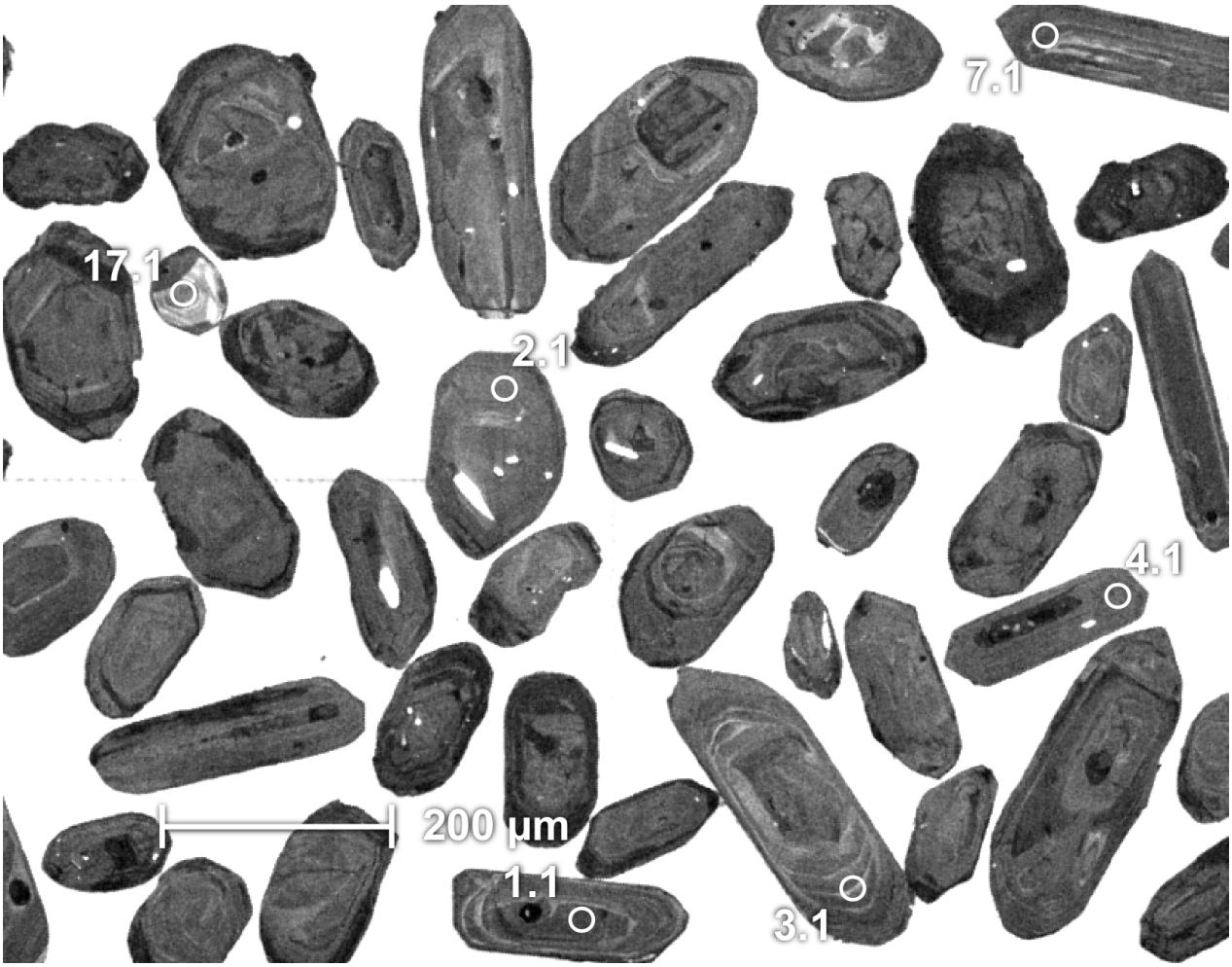

Figure 2. Cathodoluminescence image of representative zircons from sample 242442: feldspathic gneiss, Passchendeele farm. Numbered circles indicate the approximate locations of analysis sites

Analytical details

This sample was analysed on 15–16 October 2021, using SHRIMP-B. Ten analyses of the M257 standard obtained during the session indicated an external spot-to spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.28% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Eighteen analyses were obtained from 18 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). One analysis is >5% discordant. The date obtained from this analysis (Group D; Table 1) is unreliable and is considered not to be geologically significant. The remaining 17 analyses form two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 13 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2939 ± 5 Ma (MSWD = 1.2).

Group P comprises four analyses (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2915–2851 Ma.

The date of 2939 ± 5 Ma for the 13 analyses in Group I is interpreted as the magmatic crystallization age of the granodiorite. The dates of 2915–2851 Ma for the four analyses in Group P are interpreted to reflect ancient loss of radiogenic Pb.

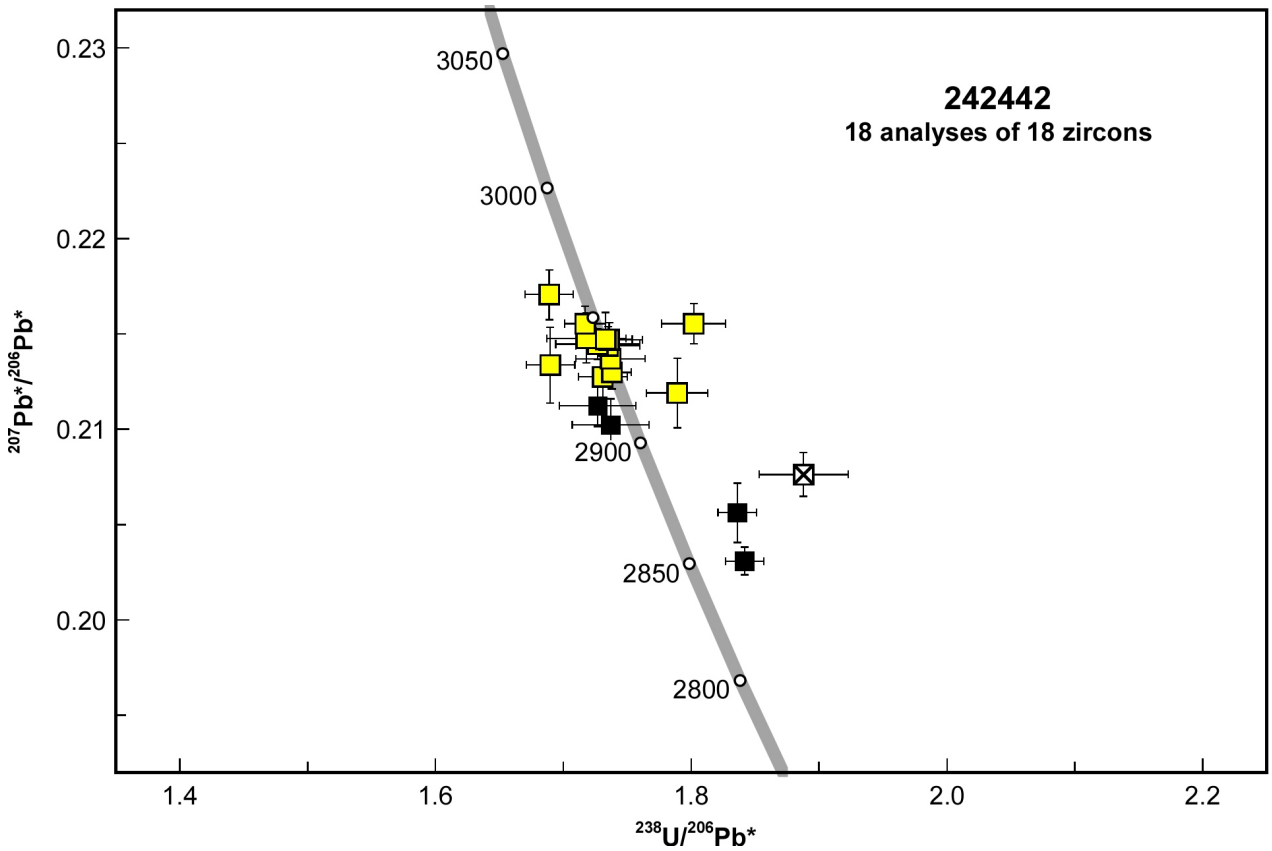

Figure 3. U–Pb analytical data for zircons from sample 242442: feldspathic gneiss, Passchendeele farm. Yellow squares indicate Group I (magmatic zircons); black squares indicate Group P (radiogenic-Pb loss); crossed square indicates Group D (discordance >5%)

Links

Table 1: Ion microprobe analytical results for zircons from sample 242442: feldspathic gneiss, Passendeele farm: 242442.1.Table1.xls
SHRIMP spot images: 242442_spots.pdf
Introduction to geochronology information: Intro_2023.pdf

References

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Wingate, MTD, Fielding, IOH, Lu, Y and Ivanic, T 2021a, 242652: felsic metavolcanic rock, Bowgarder Hills; Geochronology Record 1807: Geological Survey of Western Australia, <www.dmirs.wa.gov.au/geochron>. View Reference online

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Recommended reference for this publication

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Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

WAROX is GSWA's field observation and sample database. WAROX site IDs have the format 'ABCXXXnnnnnnSS', where ABC = geologist username, XXX = project or map code, nnnnnn = 6-digit site number, and SS = optional alphabetic suffix (maximum 2 characters).

Isotope and element analyses are routinely conducted using the GeoHistory laser ablation ICP-MS and Sensitive High-Resolution Ion Microprobe (SHRIMP) facilities at the John de Laeter Centre (JdLC), Curtin University, with the financial support of the Australian Research Council and AuScope National Collaborative Research Infrastructure Strategy (NCRIS), and the Microscopy Australia laser ablation ICP-MS facility at the Centre for Microscopy, Characterisation & Analysis (CMCA), The University of Western Australia (UWA). The TESCAN Integrated Mineral Analyser (TIMA) instrument was funded by a grant from the Australian Research Council (LE140100150) and is operated by the JdLC with the support of the Geological Survey of Western Australia, UWA and Murdoch University. Mineral analyses are routinely obtained using the electron probe microanalyser (EPMA) facilities at the CMCA at UWA, at Adelaide Microscopy at the University of Adelaide and the Central Science Laboratory at the University of Tasmania.

Digital data related to WA Geology Online, including geochronology and digital geology, are available online at the Department's Data and Software Centre and may be viewed in map context at GeoVIEW.WA. For further information about this Geochronology Record, please contact geochronology@dmirs.wa.gov.au and quote the geochronology record and/or the sample number.

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The Government of Western Australia acknowledges Aboriginal people as the traditional custodians throughout Western Australia and their enduring connection to the land, waterways and community. We pay our respects to Elders past and present.

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